

Certification Exam Prep Workshop

A Partner Learning Services offering



Professional Machine Learning
Engineer Learner Handout

For any questions, write to cloud-partner-training@google.com

Certification Exam Prep Pilot Workshops

A Partner Learning Services offering

Professional Cloud Architect
Learner Handout



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Technical Review

- [Sample Questions](#)
- [Exam Guide](#)
- [Learning Path](#)

Checklist Item	Strategy to Master
Exam guide Source: Professional Machine Learning Engineer Exam Guide	☐ Understand the exam guide's scope; unlike other exams, PMLE focuses heavily on the "Why" and "How" of model lifecycle management, not just model building. ☐ Orchestration is Key: Section 5 (Automating & Orchestrating ML Pipelines) is ~22% of the exam. Do not overlook Vertex AI Pipelines (Kubeflow/TFX). ☐ GenAI Update: The exam now explicitly covers Generative AI, Model Garden, and Vertex AI Agent Builder. ☐ ML Crash Course: Ensure you have a solid grasp of foundational ML concepts (e.g., loss functions, gradient descent) if you are newer to the field. ☐ Architecture Framework: Familiarize yourself with the "AI and Machine Learning in the Google Cloud Architecture Framework" for best practices. ☐ Practice: Take the official sample exam and review your answers to understand the question style.
Architecting Low-Code & GenAI Solution Focus: Rapid Development vs. Custom Control	 Scenario: <i>Launching a support chatbot quickly (3 weeks) using internal PDFs.</i> ☐ Vertex AI Studio (Buy) vs. Custom Training (Build): Understanding when to use pre-trained Gemini models vs. training your own. ○ Vertex AI Platform Overview ○ BUY: Prioritize managed services like Vertex AI Model Garden, Cloud ML APIs, and AutoML when timelines are tight or teams are small. ○ BUILD: Reserve custom training for niche, high-value use cases requiring deep domain adaptation, such as medical imaging. ☐ Rapid GenAI Deployment: Utilize Vertex AI Search and Conversation tools with Agent Builder to quickly deploy a generative AI solution grounded in enterprise data without manual vector database setup. ○ Vertex AI Agent Builder Product Page ☐ BigQuery ML & AutoML: Understand how to use these <i>low-code</i> services for common classification, regression, and time-series problems without needing a custom training loop.

Data Engineering & Feature Management

Focus:
Minimizing the Training-Serving Skew

- 📌 **Scenario:** Real-time recommendation engine is facing data inconsistency.
- ☐ **Data Processing:** Dataflow (Batch/Stream) vs. Dataproc (Legacy Spark/Hadoop).
- ☐ **Batch vs. Stream:** Understand the flow of batch-processed historical data (BigQuery) vs. real-time streams (Pub/Sub).
- ☐ **Training-Serving Skew:** Occurs when logic/data used for training differs from what is available at prediction time.
- ☐ **Vertex AI Feature Store:**
 - The definitive solution to mitigate training-serving skew.
 - A centralized repository to serve features with point-in-time correctness for training and low-latency for online serving.
 - [Handling Training-Serving Skew with Feature Store](#)
 - **Vertex AI Feature Store:** Serve features online directly from BigQuery data sources.
 - [About Vertex AI Feature Store | Google Cloud Documentation](#)
 - **Vertex AI Feature Store (Legacy):** Note the difference! Offers "Online Store" (for low-latency serving) and "Offline Store" (for training) architecture to ensure data consistency.
 - [Introduction to Vertex AI Feature Store \(Legacy\)](#)

Model Operations (MLOps) & Debugging

Focus:
Troubleshooting & Scaling Training Jobs

- 📌 **Scenario:** Training job failures (OOM errors) and debugging efficiency.
- ☐ **Orchestrating Custom Training:** Understanding the workflow for running custom containers and Python scripts using Vertex AI Pipelines (Kubeflow/TFX).
 - [Custom Training Overview](#)
- ☐ **Optimization Techniques:** Know how to resolve memory issues without just throwing more hardware at the problem (e.g., reduce **batch size**, enable **gradient accumulation**).
- ☐ **Scale & Hardware:** Understand when to switch from single-machine training to **distributed training**, and how to choose the appropriate hardware (**CPU, GPU, TPU**) based on the model and required throughput.
- ☐ **Distributed Training:** Apply the appropriate distribution strategy when utilizing multiple GPUs/TPUs.
- ☐ **Troubleshooting Training Jobs:** Learn how to debug common failures like ResourceExhaustedError by adjusting batch sizes or using gradient accumulation.
 - [7 Tips for Trouble-Free ML Model Training \(Google Cloud Blog\)](#)
- ☐ **Cloud Logging:** The first step in observability. Use the "Explain this log entry" (Gemini integration) to quickly diagnose cryptic errors like OOM (Out of Memory).

Model Observability <i>Focus:</i> Monitoring & Drift/Skew	 Scenario: Model performance degradation after deployment. ☐ Serving Architecture: Know the difference between Batch Inference (e.g., Dataflow, BigQuery ML) and Online Inference (e.g., Vertex AI Endpoints) and when to use each. ○ Online Prediction (low latency) vs. ○ Batch Prediction (high throughput). ☐ Testing: A/B testing and Canary deployments for safe rollouts. ☐ Vertex AI Model Monitoring: How to automatically detect when your production data diverges from your training data. ○ Introduction to Vertex AI Model Monitoring ○ Monitor Feature Skew and Drift ☐ Prediction Drift: Changes in the model's output distribution over time. ☐ Feature Skew: Changes in the input feature distribution (e.g., seasonality, user behavior changes). ☐ Continuous Training: Model alerts should trigger retraining pipelines or human review, not just logging.
Security & Governance <i>Focus:</i> Data Exfiltration & Responsible AI	 Scenario: Preventing Data Scientists from downloading sensitive PII to local laptops. ☐ IAM: Controls <i>who</i> (Identity) can access resources, while VPC-SC controls <i>where</i> resources can be accessed from. Understand the separation of duties (e.g., Data Scientist vs. MLOps Engineer roles) ☐ Responsible AI: Be familiar with assessing and mitigating risks related to fairness and bias in model training and monitoring. ☐ VPC Service Controls (VPC-SC): How to create a "service perimeter" to stop data from moving to unauthorized networks (like a laptop or public internet). ○ VPC Service Controls Overview ☐ Securing Vertex AI Workbench: Best practices for deploying User-Managed Notebooks without public IP addresses to enhance security. ○ Create a Secure Vertex AI Workbench Instance

Need an Exam Discount

Requirements to secure an exam discount:

1. Learning activities: Complete the learning requirements: <https://goo.gle/cep-info>.

If you completed the requirements more than 90 days ago, refresh your status by completing a single activity from the list again.

2. Email alignment: Your professional email address must consistently match across all platforms below:

- **Skills for Partners:** Ensure you are using the correct profile to complete your learning - set up instructions: goo.gle/SP-ACCESS.
- **CM Connect:** Ensure your profile is active and lists the same email in the "Work / Professional Email" section - linking instructions: goo.gle/cm-email-update.
- **DRP Profile:** Ensure you have a profile on Delivery Readiness Portal using the same email address - set up instructions: goo.gle/cm-email-update.

Recertifying? You can skip the learning requirements and request a 100% exam discount through the [Partner Recertification Program](#). To qualify, you must be within your eligibility window and have your partner-affiliated email linked to your certification profile. Linking *instructions* [here](#).

If your organization's quota is reached before you complete the requirements below, you'll receive a notification from cloud-partner-training@google.com.

Exam Registration

- All exams are booked via CM Connect, Google's Candidate Management Portal: <https://goo.gle/book>
 - Already have a CM Connect account: simply login.
 - New to CM Connect but have taken a Google Cloud exam: use the email address you used in Webassessor and the Forgot Password option.
 - No longer have access to the email address used in your Webassessor account: register as a new candidate and use the "Request Merge" feature to consolidate your accounts.
- Confirmed time zone accuracy for exam location. You can choose a 12 or 24-hour format.
- Choose your testing method:
 - In Person at a Test Center - take the exam at a physical testing center.

- Online with OnVUE - take the exam online from your home or office
- Regardless of exam delivery - at a test center or online - you will need to show a [government-issued ID](#) that includes your legal name, photograph, and signature. **If the name on your ID does not match the name in your CM Connect account, you will not be able to take the exam and will forfeit your entire exam fee, regardless of any discounts applied.**

Exam Accommodation

Testing accommodations are considered on a case-by-case basis and may require supporting documentation. Depending on the request, it can take up to 30 days. Additional details [here](#).

Checklist: Remote Proctored Exams

72 hours prior to your exam, ensure you meet all the requirements for [online proctored exams](#)

- ☐ Review the [online testing requirements](#) for technology, testing space, ID and testing rules
- ☐ Ensure you have downloaded all necessary software and confirmed administrative rights.
- ☐ Ensure you have a stable internet connection and a working camera and microphone.
- ☐ Ensure your video and audio quality do not interfere with effective proctoring. Issues such as loss of feed, poor resolution, excessive background noise, or audio dropouts may result in the termination of your exam.
- ☐ Ensure you have a private, quiet, well-lit room with doors where no one else will enter.
- ☐ Ensure the walls do not have any whiteboards or posters with text/diagrams.
- ☐ Ensure you have a cleared workspace free from clutter, paper, pens, or extra devices.
- ☐ Ensure you put away external monitors and close all other applications on your computer.
- ☐ Ensure you are able to show your entire testing environment (using a mobile device).
- ☐ Ensure you will not be interrupted or disturbed and no one will disturb you
- ☐ Ensure you are prepared to sit for the entire exam; you cannot step away from the camera.
- ☐ Ensure there are not unauthorized persons present in the secure Exam area
- ☐ Ensure you have reviewed [Important Program Policies](#) and understand the prohibited behaviors.

*** Consider an onsite exam if you are unable to meet any of the remote testing requirements listed above.**

Checklist: Onsite Proctored Exams

Ensure you are ready for your onsite proctored exam

- ☐ Arrive at the test center 15 minutes before your scheduled start time. Arriving more than 15 minutes late may result in being turned away and losing your fee.
- ☐ If you need to cancel or reschedule, you must do so at least 24 hours before your start time.
- ☐ Bring a [government-issued ID](#) that includes your name, photograph, and signature.

Please ensure your legal name on your CM Connect profile matches your government-issued ID. If there is a mismatch, you will be barred from the exam and will forfeit your entire fee. If you need to update your legal name, please contact [Google Cloud Certification Support](#). Submit this request at least one week before your exam to allow for processing.

- ☐ Bring an additional ID [example: credit card] that might be required for verification.
- ☐ Ensure your CM profile name matches your physical IDs exactly to avoid being turned away.
- ☐ Testing centers are often kept at a cold temperature, so dress accordingly.

Important: Remote Proctored Exams

For Remote Proctored exams, please consider that you must...

- ☐ **...have** administrative rights on your device to download all necessary software.
- ☐ **...have** a stable internet connection and a working camera and microphone.
- ☐ **...have** a private, quiet, well-lit room with a closed door and no interruptions.
- ☐ **...have** a cleared workspace free from clutter, paper, pens or extra devices.
- ☐ **...show** the proctor your entire testing environment (using a mobile device)
- ☐ **...sit** for the entire exam; no breaks and you cannot step away from the camera
- ☐ **...disconnect** external monitors and close all other applications on your computer

* Consider an onsite exam if you are unable to meet any of the remote testing requirements listed above.

Exam Day: Mental & Physical State

Maintain your focus and clarity by following these guidelines:

- ☐ **No New Material:** Avoid cramming. This prevents information overload and confusion.
- ☐ **Focus on High-Level Review:** Summary notes, flashcards, or mind maps. Focus on connecting concepts and reinforcing your existing knowledge, not deep technical dives.
- ☐ **Get Adequate Sleep:** Prioritize a full, restful night's sleep before the exam. Mental fatigue severely impacts critical thinking and recall.
- ☐ **Plan Meals & Hydration:** Have a balanced meal before the exam for sustained energy. Have water nearby (if allowed) and ensure you are not overly hungry or thirsty.
- ☐ **Stress Reduction Practice:** Practice deep breathing or a brief meditation/walk the morning of the exam. Enter the test calm and focused.

Exam Day: Test-Taking Strategy

Set yourself up for success by adopting these test-taking habits:

- ☐ **Time Management:** Know the total number of questions and the time limit. Roughly allocate a

time per question and stick to it (e.g., if 60 questions in 120 min, aim for 2 min/question).

- ☐ **Mark for Review:** Immediately flag any question that requires more than a minute to read and understand. Answer your best guess and move on. Return to marked questions later.
- ☐ **Read the Full Question & Answers:** Train yourself to look for keywords like "most cost-effective," "least latency," "highly available," or "secure." Understand the the explicit ask.
- ☐ **Elimination Strategy:** For tough, multiple-choice questions, focus on eliminating incorrect options first. This can raise your probability of selecting the correct answer from 25% to 50%.
- ☐ **Review and Final Check:** Save time at the end to review all your flagged questions and ensure you've clicked an answer for every single question before final submission.

After the exam

- An onscreen PASS/FAIL confirmation is displayed shortly after clicking the Submit button. A final confirmation is emailed after the exam within a few days
- You can check exam pass/fail status on [CM Connect](#) within 24-48 hours by accessing the "Exam History" page from the left hand navigation bar

You got this!

Questions: cloud-partner-training@google.com